

Comprehensive Database Systems MCQ Exam

Section 1: Midterm Questions (Set 1)

1. A self-join is typically used to:
 - Copy data from one table to another
 - Combine data from two unrelated tables
 - Backup a database
 - Compare rows within the same table
2. Is this statement true or false? “In SQL, NULL values represent missing or unknown data.”
 - True
 - False
3. What does the SQL command LIMIT 10 do?
 - Limits the result set to the first 10 rows
 - Selects the first 10 tables in a database
 - Limits the number of characters in the result set
 - Limits the table rows before a selection (not projection) to 10
4. Which of the following data integrity constraints ensures values in one table correspond to values in another table?
 - Foreign key constraint
 - Default constraint
 - Unique constraint
 - Check constraint
5. Which of the following SQL clauses is best suited for filtering out duplicates based on specific columns, and not all columns?
 - DISTINCT
 - ORDER BY
 - UNIQUE BY
 - GROUP BY

6. Which type of subquery is executed once for every row processed by the outer query?
- Uncorrelated Subquery
 - None of the above
 - Correlated Subquery
 - Nested Subquery
7. What distinguishes a primary key from a foreign key?
- Both a primary key and a foreign key identify a unique row in a table.
 - A primary key identifies a unique row in a table, while a foreign key links data between tables.
 - A primary key links data between tables, while a foreign key identifies a unique row in a table.
 - Both a primary key and a foreign key link data between tables.
8. Consider the two relations R and S. What would be the number of rows returned when we do a natural join on R and S?

R			S		
A	B	C	A	B	C
0	10	1000	0	102	1000
1	101	1001	1	102	1001
2	102	1002	2	102	1002

- 1
 - 3
 - 0
 - 9
9. In an `INNER JOIN`, if the `ON` condition doesn't match any rows from the tables:
- The result set will show rows from the second table.
 - The result set will have `NULL` values.
 - The result set will show rows from the first table.

- The result set will have zero rows.
10. Which of the following is NOT an aggregate function in SQL?
- COUNT
 - SUM
 - AVG
 - CONCAT
11. Is the following statement true or false? “SQL keywords are case-sensitive.”
- False
 - True
12. In SQL, the EXCEPT operator:
- Returns all the records from the second query that are not in the first query
 - Combines records from both queries
 - Returns all the records from both queries
 - Returns all the records from the first query that are not in the second query
13. Which of the following allows you to uniquely identify a tuple?
- Schema
 - Super Key
 - None of the Above
 - Domain
14. Which type of join returns only the rows where there is a match in both tables?
- INNER JOIN
 - LEFT JOIN
 - RIGHT JOIN
 - FULL JOIN
15. What does a Common Table Expression (CTE) allow you to do?

- Create a new table from an existing table
- Connect to a different database
- Modify the structure of an existing table
- Define a temporary result set that can be referred to within a SELECT, INSERT, UPDATE, or DELETE statement

16. What term below describes how all the data is organized in a database?

- tuple
- schema
- instance
- table

17. Consider the schema shown below. What SQLite error will be thrown when trying to insert the following data?

```
CREATE TABLE PERSON (
    ID char(10) primary key,
    name char(40) NOT NULL,
    father char(10),
    foreign key (father) references PERSON(ID)
);
```

```
INSERT INTO PERSON VALUES
('P001', 'Ron', NULL),
('P002', 'Sam', 'P001'),
('P003', 'Pamela', 'P006'),
('P004', 'Aron', 'P002'),
('P005', 'Jack', 'P004');
```

- FOREIGN KEY constraint failed
- None of these
- PRIMARY KEY constraint failed
- NOT NULL constraint failed

18. In a LEFT JOIN, the table placed on the left side of the JOIN keyword will have all its rows in the result, even if there's no match in the right table.

- True
- False

19. What is the result of the SQL operation `SELECT 5/2` in SQLite3?

- An error due to data type mismatch.
- 2.5
- 2
- 3

20. Which SQL keyword is used to define a condition that the result set must satisfy?

- WHERE
- GROUP
- ORDER
- JOIN

21. Consider the following tables. What will be the output of the query?

Shop Table (3 rows where `shop_id = 1`):

shop_id	item_id	quantity
1	101	100
1	103	200
1	104	120

Item Table (4 rows total):

item_id	name	price
101	milk	8.12
102	chocolate	12.99
103	bread	3.45
104	eggs	10

Query:

```
SELECT COUNT(*) FROM Shop, Item WHERE shop_id = 1;
```

- 36
- 4
- 3

- 12

22. Consider a `LEFT OUTER JOIN` between tables A and B. If there's a `WHERE` condition that filters out `NULL`s for columns originating from table B:

- The join type changes to a `RIGHT OUTER JOIN`.
- The join effectively becomes an `INNER JOIN`.
- The join remains unaffected, and will still return all rows from A.
- The result will be an empty set if all records from B are `NULL` for the join condition.

23. Consider `Department(dept_id, name)` with 5 rows and `Employee(emp_id, name, dept_id)` with 100 rows. What is the maximum number of rows returned by the query?

```
SELECT * FROM Department as d
  LEFT OUTER JOIN Employee as e
    ON d.dept_id = e.dept_id;
```

- 105
- 500
- 104
- 5

24. Monotonic queries are:

- Always increasing in value
- Queries that produce results that only stay the same or increase in size
- Queries that include both existential and universal quantifiers
- Queries that include only universal quantifiers

25. Which of the following statements is true regarding a SQL query containing multiple uncorrelated subqueries?

- The outermost subquery executes first, followed by its inner subquery, and so on.

- The innermost subquery executes first, followed by its parent, and so on.
 - Subqueries cannot be nested more than two levels deep.
 - All subqueries execute simultaneously with the main query.
26. For SQLite3 what would be the outcome of the SQL operation `SELECT '100' + 10`?
- '10010'
 - An error due to data type mismatch.
 - 110
 - '110'
27. What would be the result of the following SQL function call: `ROUND(123.4567, 2)`?
- 123.45
 - 123
 - 124
 - 123.46
28. Assuming a `FULL OUTER JOIN` between `employees(salary)` and `departments(budget)`, which statements are true?
- (1) Rows with employees who have no department will have NULL in the `budget` column.
 - (2) Rows with departments that have no employees will have NULL in the `salary` column.
- Both statements are correct
 - Neither statements are correct
 - Only statement 2 is correct
 - Only statement 1 is correct
29. Consider a `LEFT JOIN` between tables A (50 records) and B (30 records). What is the maximum number of rows that the result can have?
- 30
 - 80
 - 1500

- 50

30. When using a **CASE** statement in SQL, which keyword provides an alternative result if none of the conditions are met?

- **DEFAULT**
- **ELSE**
- **OTHERWISE**
- **ELSEIF**

Section 2: Final Exam Questions (Set 2)

31. Relational algebra is a ----- language used to query and operate on relations.

- Declarative
- Functional
- Object-Oriented
- Procedural

32. Clustered indexes in a database:

- Cannot be created in SQL
- Do not store data physically in index order
- Store data physically in the order of the index
- Are always less efficient than unclustered indexes

33. Extended Relational Algebra is based on multi-set theory and SQL is based on set theory.

- True
- False

34. What is the relational algebra equivalent of the SQL query:

```
SELECT * FROM Orders
WHERE orderdate BETWEEN '2023-01-01' AND '2023-12-31'
```


- σ (orderdate $\geq$ '2023-01-01' AND orderdate $\leq$ '2023-12-31') (π orderdate(Orders))
 - σ orderdate BETWEEN '2023-01-01' AND '2023-12-31' (Orders)
 - σ (orderdate $\geq$ '2023-01-01' AND orderdate $\leq$ '2023-12-31') (Orders)
 - σ (orderdate $\geq$ '2023-01-01' AND orderdate $\leq$ '2023-12-31') (Orders)
35. 3NF states that each non-key attribute must depend on "the key, the whole key, and nothing but the key."
- True
 - False
36. Choose the incorrect statement below:
- A Super Key is always a Candidate Key
 - A Foreign Key can be a Super Key
 - A Primary Key is always a Super Key
 - A Candidate Key is always a Super Key
37. A primary key must always come from a candidate key.
- False
 - True
38. Given the relation R(ABC) and functional dependencies $\{A \rightarrow B, B \rightarrow C\}$. The attribute A is a superkey, but not a candidate key.
- True
 - False
39. In a B-tree of order m, which of the following statements is false?
- The root has at least two children if it is not a leaf node
 - All data pointers appear at the leaf level
 - Every node has at most m children
 - A non-leaf node with k children contains k - 1 keys

40. Given an initially empty B-tree with degree $n=3$, insert the keys 20, 10, 25, 15, 5, 30, 35 in sequence. Is the B-tree shown in the accompanying image correct?

- True
- False

41. How can the following query be 'unnested' to improve performance?

```
SELECT name
  FROM Students
 WHERE student_id IN (
    SELECT student_id
      FROM Enrollments
    WHERE course_id = 'DSC100');
```

- Replace the subquery with a JOIN operation.
- Increase the search condition in the WHERE clause.
- Eliminate the WHERE clause.
- Use an EXISTS clause instead of IN.

42. Which of the following best describes the use of universal quantifiers (ALL) compared to existential quantifiers (ANY or SOME)?

- ALL is used only in DELETE statements, while ANY is used in SELECT statements.
- ALL and ANY are functionally identical and can be used interchangeably.
- ALL is less efficient than ANY in all scenarios.
- ALL checks if all elements in a set meet a condition, while ANY or SOME check if at least one element meets the condition.

43. How is data typically stored on a traditional hard disk drive (HDD)?

- Data is stored using optical technology, similar to how it's stored in a CD or DVD.
- Data is stored in sectors and tracks on a spinning magnetic disk.
- Data is always stored in sequential blocks that are uniformly sized for all types of data.

- Data is always stored in a random access pattern, similar to how it's stored in RAM.
44. Consider a database table with a non-clustered index on one column and a clustered index on another. How does the presence of these indexes affect the performance of `UPDATE` operations on the table, and why?
- Clustered indexes have no impact on `UPDATE` operations.
 - `UPDATE` operations are slower due to the overhead of maintaining both indexes.
 - `UPDATE` performance varies depending on whether the operation modifies indexed columns or non-indexed columns.
 - Non-clustered indexes improve `UPDATE` performance regardless of the columns involved.
45. In what order does the SQL engine logically process the clauses of this query?

```
SELECT m.id, AVG(e.salary)
  FROM Employees AS e
  JOIN Managers AS m ON e.manager_id = m.id
 WHERE e.department_id = 3
  GROUP BY m.id
HAVING AVG(e.salary) > 40;
```

- `SELECT` → `FROM` → `JOIN` → `WHERE` → `GROUP BY` → `HAVING`
 - `FROM` → `JOIN` → `WHERE` → `GROUP BY` → `HAVING` → `SELECT`
 - `WHERE` → `FROM` → `JOIN` → `SELECT` → `HAVING` → `GROUP BY`
 - `FROM` → `WHERE` → `JOIN` → `GROUP BY` → `HAVING` → `SELECT`
46. Consider a table `OrderDetails` with PK (`OrderID`, `ProductID`) and FDs: `ProductID` → `SupplierID`, `SupplierID` → `SupplierName`. Which statement is true?
- The table is in 2NF as it has a composite primary key, and all non-key attributes are functionally dependent on it.
 - The table is in 3NF as there are no transitive dependencies present in the table.

- The table is not in 2NF because of a partial dependency of **SupplierID** and **SupplierName** on a part of the primary key.
 - The table is in 2NF but not in 3NF due to a transitive dependency between **SupplierID** and **SupplierName**.
47. Which statement about "pipelining" in query execution is most true?
- Each operator processes all the data at the same time, in parallel.
 - Operations form a queue, where data is passed from one step to another.
 - Pipelining allows each operator to start processing as soon as its predecessor produces the first few tuples, without waiting for the entire operation to complete.
 - Pipelining in query execution refers to the physical transmission of data through a pipeline in the hardware layer.
48. Select the statement that most accurately describes "materialization" in query execution.
- Query materialization is a process where all query results are stored permanently in the database for future use, similar to creating a view.
 - Materialization involves temporarily persisting data between query operations, often used when immediate consumption of data is not possible or optimal.
 - In materialization, each operator must transmit its complete output to a temporary storage before the next operator can begin processing.
 - Materialization is the process of optimizing a query by breaking it down into simpler subqueries that can be executed independently.
49. Choose the statement that best describes scenarios where set operations are preferred over joins.
- Set operations are used to find combinations or differences between two datasets, especially when the queries produce the same columns and simplicity or performance is a priority.
 - Set operations are preferred when you need to create complex queries that manipulate several columns from different tables based on multiple conditions.

- You should always use set operations over joins because they are simpler and do not require the underlying tables to have any common keys or columns.
- Joins are less performant than set operations and cannot combine data from different databases, which is why set operations are generally favored in all scenarios.

50. What is the primary purpose of using the `EXPLAIN ANALYZE` statement in PostgreSQL?

- It is utilized to change the structure of the query plan cache, forcing PostgreSQL to find a new plan for subsequent executions.
- It is a debugging tool that checks the syntax of the SQL query and ensures that it adheres to the PostgreSQL standards without executing it.
- It does not execute the query, but provides a detailed execution plan, including cost, which helps in understanding and optimizing query performance.
- It executes the query and provides a detailed execution plan, including timing, which helps in understanding and optimizing query performance.

Section 3: Additional Practice Questions

51. Which statement about relations is TRUE?

- Relations must have at least one tuple
- The order of tuples in a relation matters
- All attributes must be of different types
- A relation is a set of tuples with the same attributes

52. Given a relation $R(A,B,C,D)$ with functional dependencies $A \rightarrow B$ and $B \rightarrow C$, what is TRUE?

- A is a superkey
- $\{A,D\}$ is not a superkey
- B must be a primary key
- $A \rightarrow C$ by transitivity

53. Which of the following is NOT an atomic data type in SQL?
- INTEGER
 - VARCHAR
 - ARRAY
 - DATE
54. Which statement about NULL values in SQL is correct?
- `NULL = NULL` evaluates to `TRUE`
 - NULL values are considered equal to each other
 - NULL values are ignored in `COUNT(*)`
 - NULL values are excluded from `COUNT(column_name)`
55. When is a non-clustered index LEAST useful?
- When querying a small percentage of rows
 - When the table is frequently updated
 - When performing range queries
 - When doing full table scans
56. In an ER diagram, what does a double-line (thick line) relationship indicate?
- Many-to-many relationship
 - Total participation
 - Weak entity relationship
 - Inheritance relationship